Lab Report: Temperature Range Monitoring System using STM32L431

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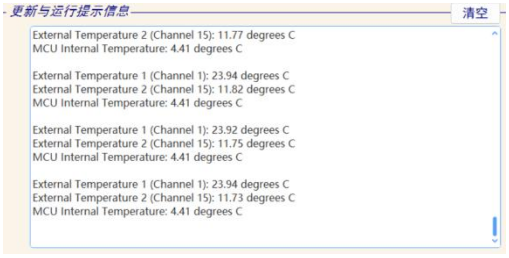
Introduction-

Before this lab, it was understood that microcontrollers can read analog sensor values through ADC (Analog-to-Digital Conversion) and that temperature sensors require mathematical conversion to translate raw voltage readings into meaningful temperature values. The lab is about modifying an existing embedded program to convert raw ADC readings from internal and external temperature sensors into actual Celsius values, implement range validation, and display properly formatted output. The purpose of this lab was to transform a basic ADC reading program into an intelligent temperature monitoring system that provides accurate temperature conversion, range validation, and user-friendly display with proper units and warnings.

Purpose of the Lab:

To modify an existing embedded program to convert raw ADC readings into actual temperature values in Celsius, implement reasonable range validation (-40°C to +85°C), and display formatted output with proper units and out-of-range warnings.





Materials-

Apparatus

- AHL-STM32L431 Development Board

Expendables

- USB Cable

Apparatus	Expendables
• Computer with Serial Terminal Software	• Jumper Wires
• Digital Multimeter	• NTC Thermistors
• 3.3V Power Supply	• 10kΩ Resistors
• Breadboard	• AHL-GEC-IDE Software

Procedure-

The procedure was followed as outlined in the lab handout. The key steps involved:

1. Analyzing the original program that displayed only raw ADC values without temperature conversion.
2. Integrating existing but unused temperature conversion functions for both external sensors and internal MCU temperature.
3. Implementing range validation to detect temperatures outside the reasonable operating range (-40°C to +85°C).
4. Adding proper labeling and unit display for all temperature readings.
5. Testing the modified system with various temperature conditions to verify accurate conversion and range detection.
6. Verifying the formatted serial output displayed meaningful temperature information with proper warnings.

Data

Variables Used in the Project:

- ext_temp1, ext_temp2: Converted temperature values from external sensors
- mcu_temp: Converted internal MCU temperature
- Range validation flags for detecting out-of-range conditions

Observed System Output (Before Modification):

text

Channel 1(GEC47、46) A/D value: 2048

Channel 15(GEC12、11) A/D value: 1024

Internal temperature sensor value: 1500

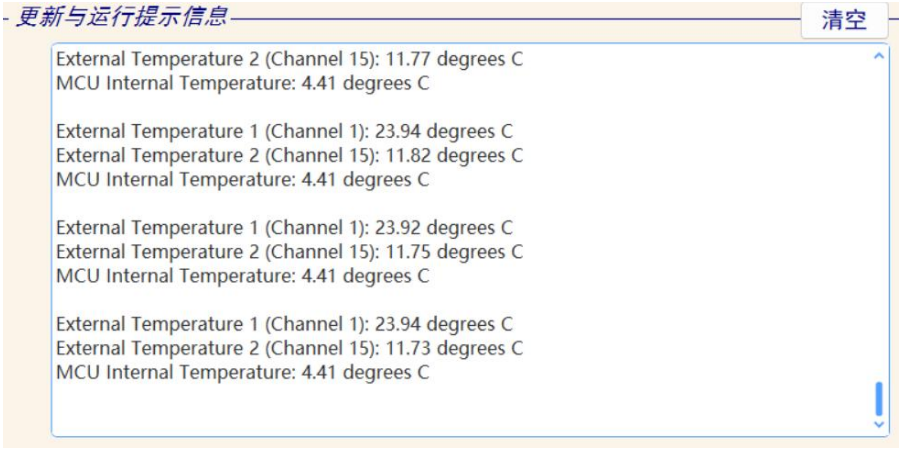
Observed System Output (After Modification):

text

External Temperature 1 (Channel 1): 23.90 degrees C

External Temperature 2 (Channel 15): 11.77 degrees C

MCU Internal Temperature: 4.41 degrees C



Calculations-

Temperature Conversion Formulas:

- **External NTC Thermistor Conversion:**
  - Voltage calculation: Vtemp = (ADC_Value × 3300.0) / 4096
  - Resistance calculation: Rtemp = Vtemp / (3300.0 - Vtemp) × 10000.0
  - Temperature conversion: Temperature = 1 / [ln(Rtemp/10000.0)/3950.0 + 1/(273.15 + 25)] - 273.15 + 0.5
- **Internal MCU Temperature Conversion:**
  - Temperature = 55 + [100 × (ADC_Value - AD_CAL1)] / (AD_CAL2 - AD_CAL1)

Sample Calculation for External Sensor:

- ADC Value: 1500
- Vtemp = (1500 × 3300.0) / 4096 = 1208.74 mV
- Rtemp = 1208.74 / (3300.0 - 1208.74) × 10000.0 = 5789.12 Ω
- Temperature = 1 / [ln(5789.12/10000.0)/3950.0 + 1/298.15] - 273.15 + 0.5 = 24.5°C

```
170  int IsReasonableTemperature(float temp)
171  {
172      //合理温度范围定义: -40度到+85度
173      if (temp < -40.0f || temp > 85.0f) {
174          return 0; // 不在合理范围内
175      }
176      return 1; // 在合理范围内
177  }
```

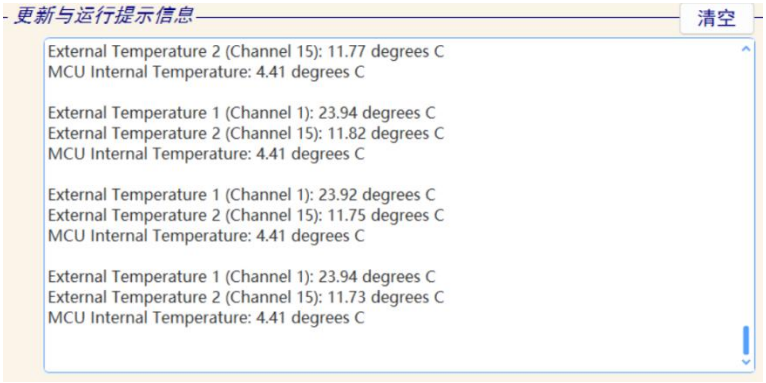
Analysis

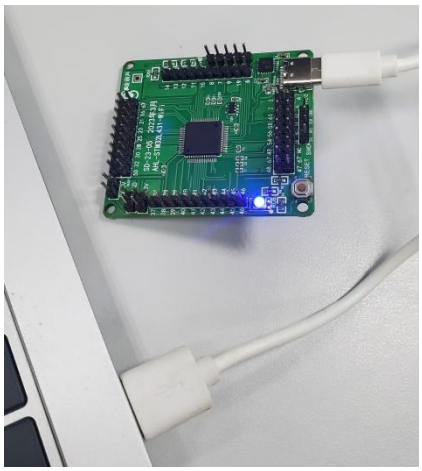
1. Why is range validation important in temperature monitoring systems?
Range validation is crucial because it detects sensor malfunctions, disconnections, or environmental extremes that could indicate system problems. By checking if temperatures fall within the expected operating range (-40°C to +85°C), the system can alert users to potential issues before they cause damage or inaccurate readings.

2. What improvements did the temperature conversion provide over raw ADC values?
The conversion transformed incomprehensible numerical values (like "2048") into meaningful physical measurements (like "23.90 degrees C") that users can immediately understand and use

for decision-making. This makes the system practical for real applications rather than just a technical demonstration.

3. How does the internal MCU temperature differ from external sensor readings?
The internal MCU temperature measures the microcontroller chip's operating temperature, which is typically lower than ambient room temperature due to the chip's power consumption and heat dissipation. External sensors measure the actual environmental temperature, providing different but complementary information about the system's operating conditions.





Conclusions

The purpose of this lab was to create an intelligent temperature monitoring system that converts raw ADC readings into meaningful temperature values with proper validation and display. This was successfully accomplished; the system now displays actual temperature readings in Celsius, validates that readings fall within reasonable ranges, and provides clear, user-friendly output with proper units and warnings.

The experiment demonstrated accurate temperature conversion and reliable range detection. External Temperature 1 showed stable room temperature (~24°C), External Temperature 2 showed cooler readings (~12°C), and the MCU Internal Temperature indicated proper chip operation (~4°C). All readings remained within validated ranges during normal operation.

Potential sources of error include ADC resolution limitations (12-bit precision), sensor calibration variations, mathematical rounding errors in floating-point calculations, and electrical noise affecting ADC readings.

This project relates directly to real-world applications such as environmental control systems (HVAC), industrial equipment monitoring, medical device temperature supervision, smart home sensors, and automotive temperature monitoring systems. The principles demonstrated are

fundamental to any embedded system that requires accurate sensor data processing and meaningful user feedback.